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2014.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

# COOLING SYSTEM CONCENTRATION CHECK (G2154751)

## GENERAL EQUIPMENT

| EQUIPMENT NAME             |
|----------------------------|
| Refractometer / Hydrometer |

## CHECK



### CAUTIONS:

- The engine cooling system must be maintained with the correct concentration and type of anti-freeze solution to prevent corrosion and frost damage.
- Engine coolant will damage the paint finished surfaces. If spilt, immediately remove the coolant and clean the area with water.
- Never remove the coolant expansion tank cap under any circumstances while the engine is operating.



### NOTES:

- This procedure contains illustrations showing certain components removed to provide extra clarity.
- This procedure contains some variation in the illustrations depending on the vehicle specification, but the essential information is always correct.

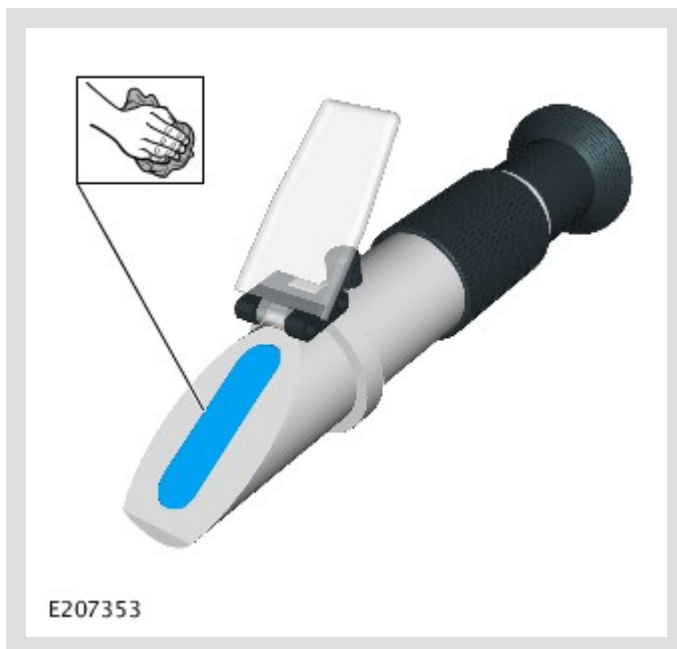


Raise the cover plate.

General Equipment: [Refractometer](#) / [Hydrometer](#)

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2.



Clean the prism surface with a soft cloth.

3.



Put 2 drops of pure distilled water on the prism surface.

4.



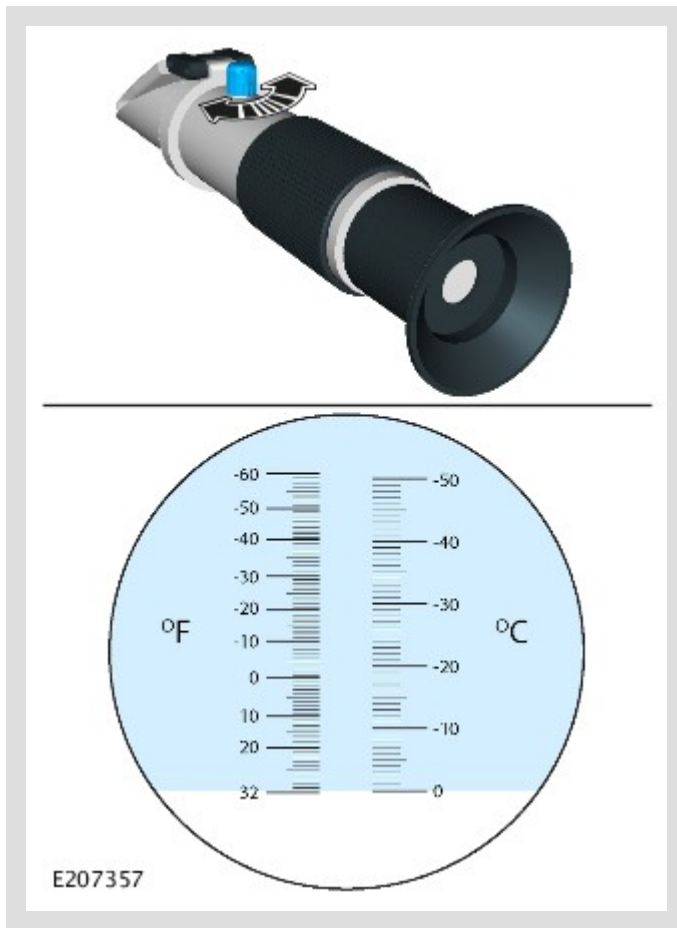
Lower the cover plate.

5.



Observe the shadow line on the scale.

6.



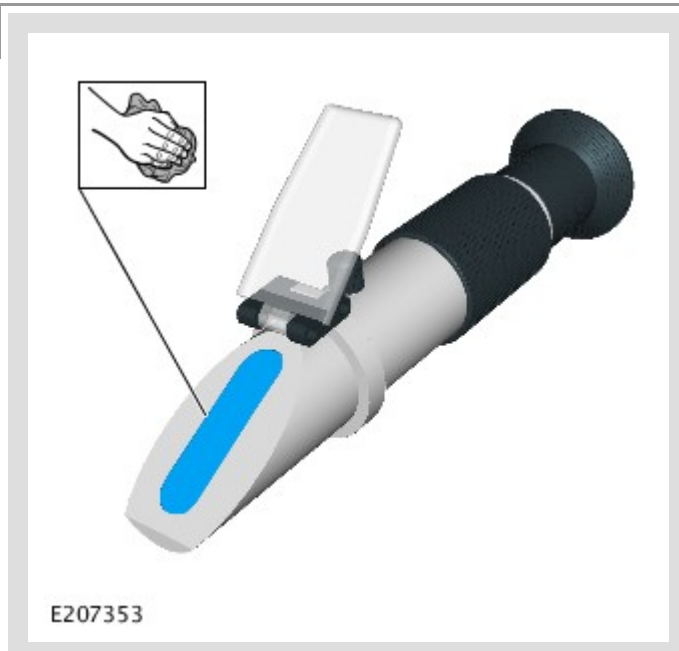
If required, use the adjustment screw to align the shadow line as shown.

7.



Raise the cover plate.

8.



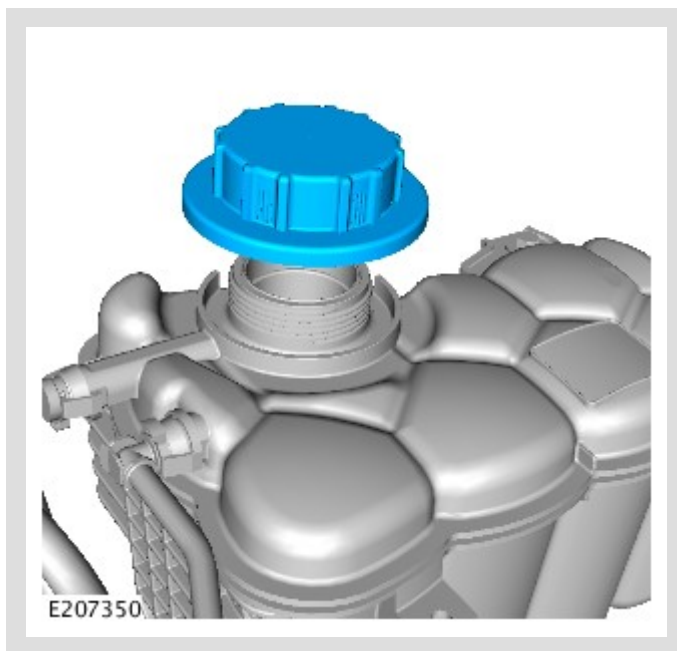
Clean the prism surface with a soft cloth.

**WARNING:**

Release the cooling system pressure by slowly turning the coolant expansion tank cap a quarter of a turn. Cover the expansion tank cap with a thick cloth to prevent the possibility of scalding.

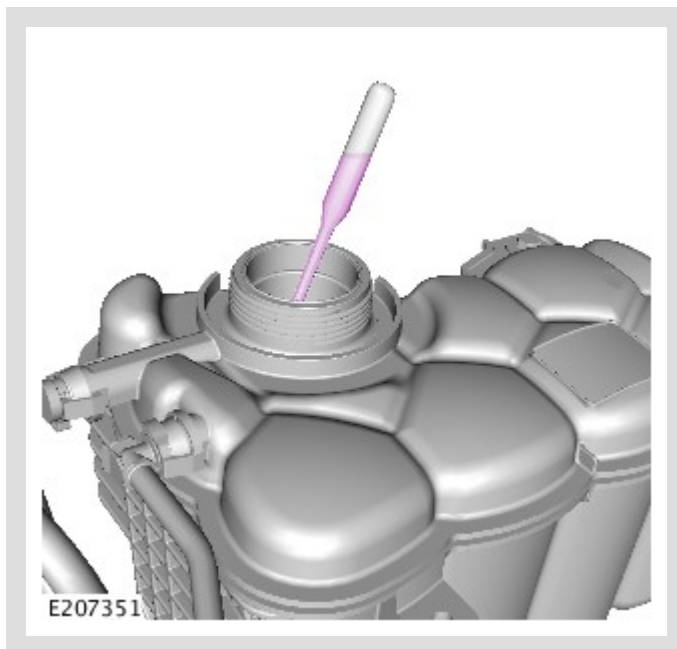
**CAUTIONS:**

- Be prepared to collect escaping coolant.
- Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.



Remove the coolant expansion tank cap.





Use the pipette to extract a coolant sample.

11.



Raise the cover plate.

12.



Put 2 drops of coolant on the prism surface.

13.

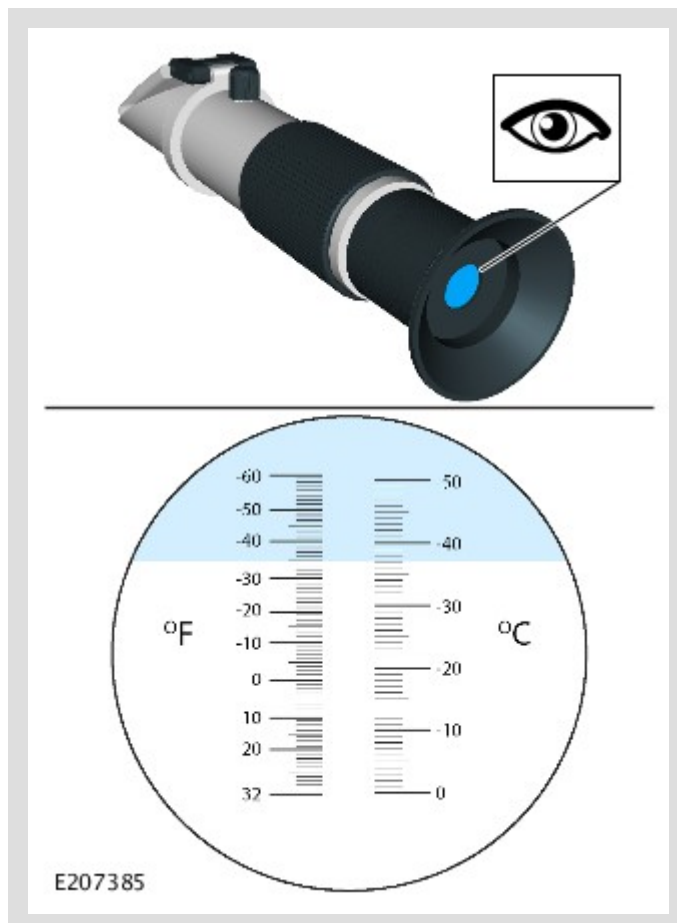


Lower the cover plate.

14.

 **NOTE:**

50% coolant concentration mix provides a freeze point of  $-37^{\circ}\text{C}$ .



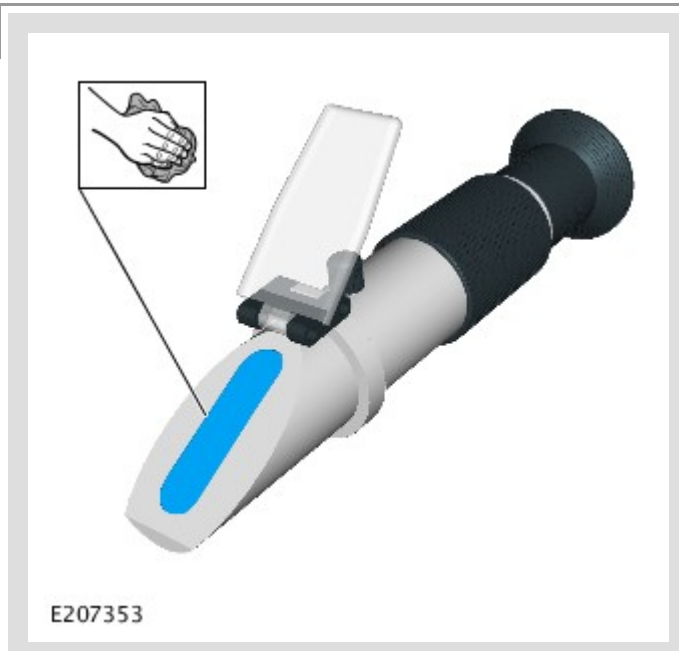
Observe the shadow line on the scale.

15.

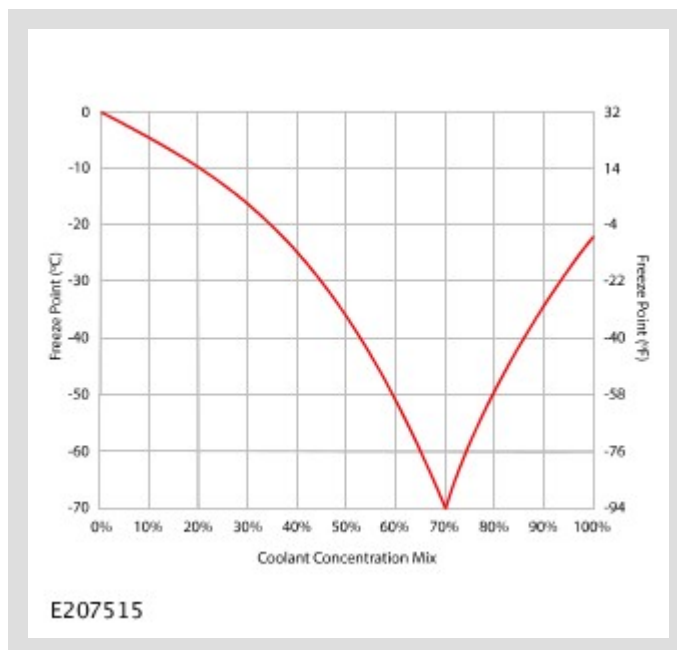


Raise the cover plate.

16.



Clean the prism surface with a soft cloth.



- If the coolant concentration mix is at 50%, proceed to Step 20.
- If the coolant concentration mix is below 50%, proceed to Step 18.
- If the coolant concentration mix is below 50%, proceed to Step 19.

**NOTE:**

Coolant concentration mix % must be divided by 100. 50% would be calculated as 0.5

- If the coolant concentration mix is below 50%, the following calculation steps must be followed.
- Step 1. Multiply the required coolant concentration mix % by the cooling system capacity.
- Step 2. Multiply the current coolant concentration mix % by the cooling system capacity.

- Step 3. Subtract the total of step 1 by total of step 2.
- Step 4. Subtract the current coolant concentration mix % from 1.
- Step 5. Divide the total of step 3 by the total of step 4.
- Drain the coolant quantity given by the calculations and replace with concentrated antifreeze.
- Proceed to Step 20.

19.

 **NOTE:**

Coolant concentration mix % must divided by 100. 50% would be calculated as 0.5

- If the coolant concentration mix is above 50%, the following calculation steps must be followed.
- Step 1. Multiply the current coolant concentration mix % by the cooling system capacity.
- Step 2. Multiply the required coolant concentration mix % by the cooling system capacity.
- Step 3. Subtract the total of step 1 by total of step 2.
- Step 5. Divide the total of step 3 by the current coolant concentration mix %.
- Drain the coolant quantity given by the calculations and replace with distilled water.

20.

 **CAUTION:**

Correct installation of the coolant expansion tank cap can be obtained by tightening the cap until 3 audible clicks are heard.

Install the coolant expansion tank pressure cap.